

S-80 Answers to Odd-Numbered Exercises

replace each 0 by **F**, 1 by **T**, Boolean sum by $\vee$, Boolean product by $\wedge$, and $\bar{}$ by $\neg$ (and x by p and y by q so that the variables look like they represent propositions, and the equals sign by the logical equivalence symbol), then $\overline{xy} = \bar{x} \vee \bar{y}$ becomes $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\overline{x+y} = \bar{x} \bar{y}$ becomes $\neg(p \vee q) \equiv \neg p \wedge \neg q$. **35.** By the domination, distributive, and identity laws, $x \vee x = (x \vee x) \wedge 1 = (x \vee x) \wedge (x \vee \bar{x}) = x \vee (x \wedge \bar{x}) = x \vee 0 = x$. Similarly, $x \wedge x = (x \wedge x) \vee 0 = (x \wedge x) \vee (x \wedge \bar{x}) = x \wedge (x \vee \bar{x}) = x \wedge 1 = x$. **37.** Because $0 \vee 1 = 1$ and $0 \wedge 1 = 0$ by the identity and commutative laws, it follows that $\bar{0} = 1$. Similarly, because $1 \vee 0 = 1$ and $1 \wedge 0 = 0$, it follows that $\bar{1} = 0$. **39.** First, note that $x \wedge 0 = 0$ and $x \vee 1 = 1$ for all x , as can easily be proved. To prove the first identity, it is sufficient to show that $(x \vee y) \vee (\bar{x} \wedge \bar{y}) = 1$ and $(x \vee y) \wedge (\bar{x} \wedge \bar{y}) = 0$. By the associative, commutative, distributive, domination, and identity laws, $(x \vee y) \vee (\bar{x} \wedge \bar{y}) = y \vee [x \vee (\bar{x} \wedge \bar{y})] = y \vee [(x \vee \bar{x}) \wedge (x \vee \bar{y})] = y \vee [1 \wedge (x \vee \bar{y})] = y \vee (x \vee \bar{y}) = (y \vee \bar{y}) \vee x = 1 \vee x = 1$ and $(x \vee y) \wedge (\bar{x} \wedge \bar{y}) = \bar{y} \wedge [x \wedge (x \vee y)] = \bar{y} \wedge [(x \wedge x) \vee (x \wedge y)] = \bar{y} \wedge [0 \vee (x \wedge y)] = \bar{y} \wedge (x \wedge y) = \bar{x} \wedge (y \wedge \bar{y}) = \bar{x} \wedge 0 = 0$. The second identity is proved in a similar way. **41.** Using the hypotheses, Exercise 35, and the distributive law it follows that $x = x \vee 0 = x \vee (x \vee y) = (x \vee x) \vee y = x \vee y = 0$. Similarly, $y = 0$. To prove the second statement, note that $x = x \wedge 1 = x \wedge (x \wedge y) = (x \wedge x) \wedge y = x \wedge y = 1$. Similarly, $y = 1$. **43.** Use Exercises 39 and 41 in the Supplementary Exercises in Chapter 9 and the definition of a complemented, distributed lattice to establish the five pairs of laws in the definition.

Section 12.2

1. a) $\bar{x} \bar{y} \bar{z}$ **b)** $\bar{x} \bar{y} \bar{z}$ **c)** $\bar{x} \bar{y} \bar{z}$ **d)** $\bar{x} \bar{y} \bar{z}$ **3. a)** $xyz + xy\bar{z} + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$ **b)** $xyz + xy\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$ **c)** $xyz + xy\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$ **d)** $\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z}$ **5.** $wxy\bar{z} + wx\bar{y}z + w\bar{x}yz + \bar{w}x\bar{y}z + \bar{w}\bar{x}yz + \bar{w}\bar{x}\bar{y}z + \bar{w}\bar{x}\bar{y}\bar{z}$ **7. a)** $\bar{x} + \bar{y} + z$ **b)** $x + y + z$ **c)** $x + \bar{y} + z$ **9.** $y_1 + y_2 + \dots + y_n = 0$ if and only if $y_i = 0$ for $i = 1, 2, \dots, n$. This holds if and only if $x_i = 0$ when $y_i = x_i$ and $x_i = 1$ when $y_i = \bar{x}_i$. **11. a)** $x + y + z$ **b)** $(x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(\bar{x} + y + z)(\bar{x} + y + \bar{z})$ **c)** $(x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(x + \bar{y} + \bar{z})$ **d)** $(x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(\bar{x} + \bar{y} + \bar{z})$ **13. a)** $x + y + z$ **b)** $x + [y + (\bar{x} + z)]$ **c)** $(x + \bar{y})$ **d)** $[x + (\bar{x} + \bar{y} + \bar{z})]$

15. a)

x	$\bar{x}$	$x \downarrow x$
1	0	0
0	1	1

b)

x	y	xy	$x \downarrow x$	$y \downarrow y$	$(x \downarrow x) \downarrow (y \downarrow y)$
1	1	1	0	0	1
1	0	0	0	1	0
0	1	0	1	0	0
0	0	0	1	1	0

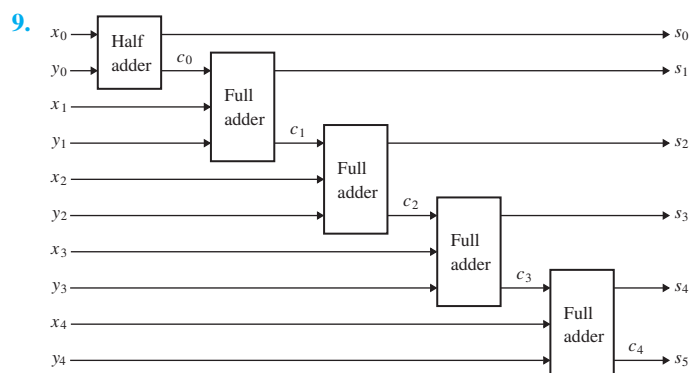
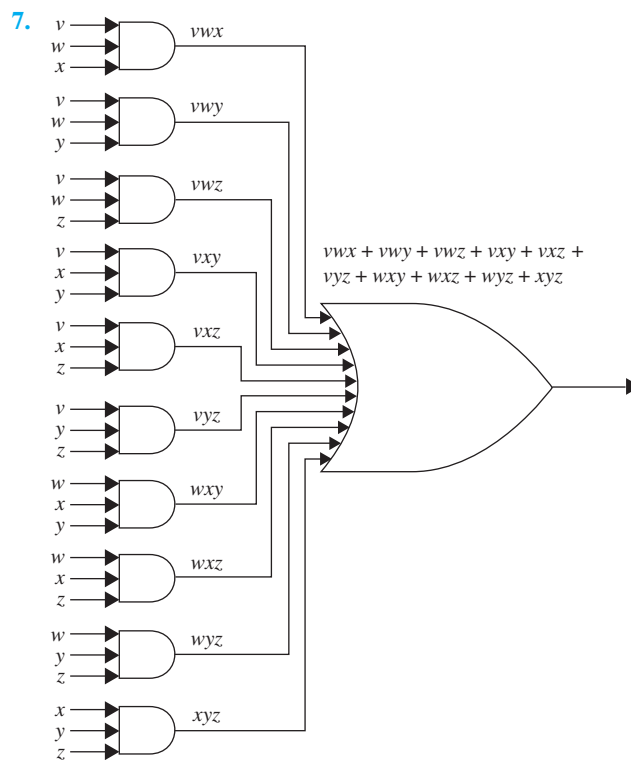
c)

x	y	$x + y$	$(x \downarrow y)$	$(x \downarrow y) \downarrow (x \downarrow y)$
1	1	1	0	1
1	0	1	0	1
0	1	1	0	1
0	0	0	1	0

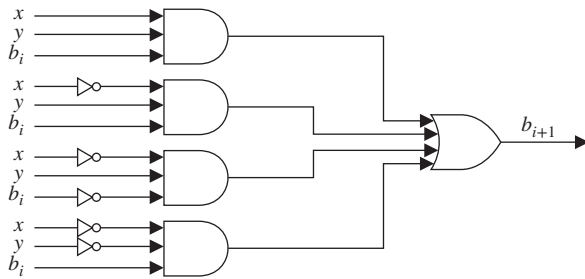
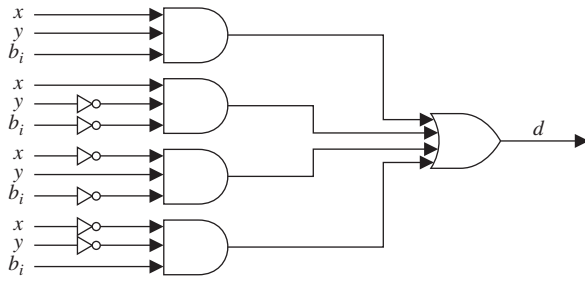
17. a) $\{[(x \mid x) \mid (y \mid y)] \mid [(x \mid x) \mid (y \mid y)]\} \mid (z \mid z)$ **b)** $\{[(x \mid x) \mid (z \mid z)] \mid y\} \mid \{[(x \mid x) \mid (z \mid z)] \mid y\}$ **c)** x **d)** $[x \mid (y \mid y)] \mid [x \mid (y \mid y)]$ **19.** It is impossible to represent $\bar{x}$ using $+$ and $\cdot$ because there is no way to get the value 0 if the input is 1.

Section 12.3

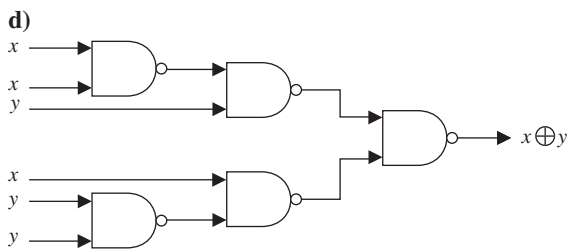
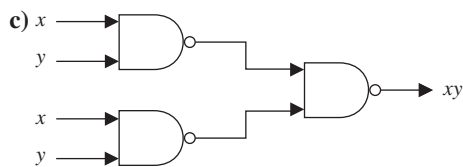
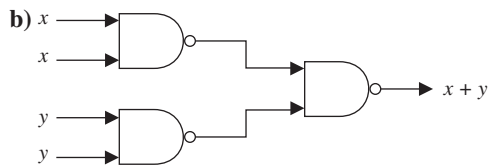
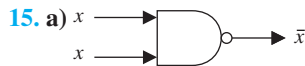
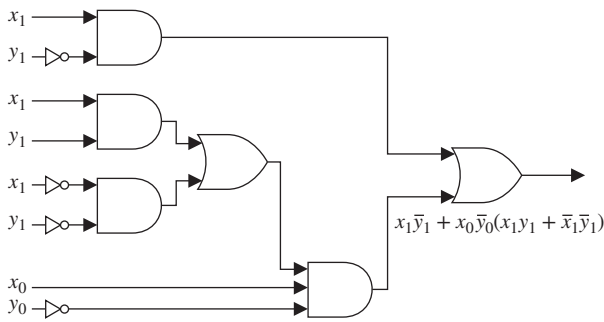
1. $(x+y)\bar{y}$ **3.** $(\bar{x}y) + (\bar{z} + x)$ **5.** $(x+y+z) + (\bar{x} + y + z) + (\bar{x} + \bar{y} + \bar{z})$



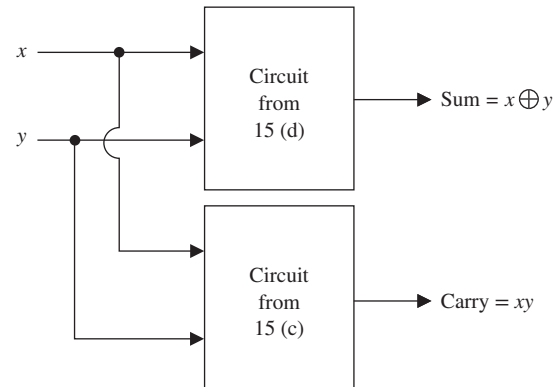
11.



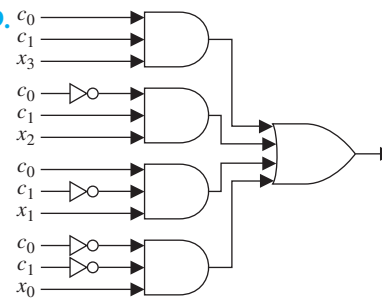
13.



17.



19.



Section 12.4

1. a) y $\bar{y}$ b) xy and $\bar{x}\bar{y}$

	y	$\bar{y}$
x		
$\bar{x}$	1	

3. a) y $\bar{y}$

	y	$\bar{y}$
x		1
$\bar{x}$		

b) y $\bar{y}$ c) y $\bar{y}$

	y	$\bar{y}$
x	1	
$\bar{x}$		1

	y	$\bar{y}$
x	1	1
$\bar{x}$	1	1

5. a) yz $y\bar{z}$ $\bar{y}\bar{z}$ $\bar{y}z$

	yz	$y\bar{z}$	$\bar{y}\bar{z}$	$\bar{y}z$
x				
$\bar{x}$		1		

b) $\bar{x}yz$, $\bar{x}\bar{y}\bar{z}$, $xy\bar{z}$

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7. a)

	yz	$y\bar{z}$	$\bar{y}z$	$\bar{y}\bar{z}$
x			1	
$\bar{x}$				

b)

	yz	$y\bar{z}$	$\bar{y}z$	$\bar{y}\bar{z}$
x				
$\bar{x}$	1		1	

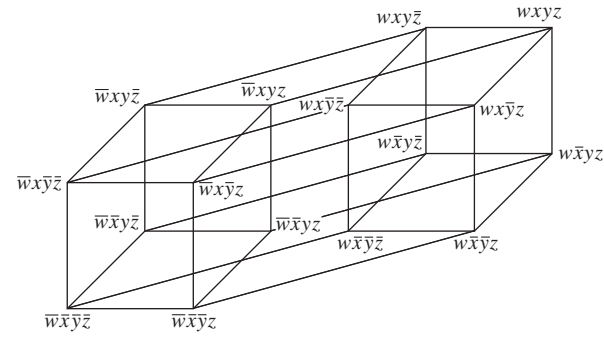
c)

	yz	$y\bar{z}$	$\bar{y}z$	$\bar{y}\bar{z}$
x	1	1		
$\bar{x}$		1		1

9. Implicants: xyz , $xy\bar{z}$, $x\bar{y}z$, $\bar{x}y\bar{z}$, xy , $x\bar{z}$, $y\bar{z}$; prime implicants: xy , $x\bar{z}$, $y\bar{z}$; essential prime implicants: xy , $x\bar{z}$, $y\bar{z}$

	yz	$y\bar{z}$	$\bar{y}z$	$\bar{y}\bar{z}$
x	1	1	1	
$\bar{x}$		1		

11. The 3-cube on the right corresponds to w ; the 3-cube given by the top surface of the whole figure represents x ; the 3-cube given by the back surface of the whole figure represents y ; the 3-cube given by the right surfaces of both the left and the right 3-cube represents z . In each case, the opposite 3-face represents the complemented literal. The 2-cube that represents wz is the right face of the 3-cube on the right; the 2-cube that represents $\bar{x}y$ is bottom rear; the 2-cube that represents $\bar{y}\bar{z}$ is front left.



13. a)

	yz	$y\bar{z}$	$\bar{y}z$	$\bar{y}\bar{z}$
wx				
$w\bar{x}$				
$\bar{w}x$				
$\bar{w}\bar{x}$		1		

b) $\bar{w}xyz$, $\bar{w}\bar{x}y\bar{z}$, $\bar{w}x\bar{y}z$, $wxy\bar{z}$

15. a)

	$x_3x_4x_5$	$x_3x_4\bar{x}_5$	$x_3\bar{x}_4x_5$	$x_3\bar{x}_4\bar{x}_5$	$\bar{x}_3x_4x_5$	$\bar{x}_3x_4\bar{x}_5$	$\bar{x}_3\bar{x}_4x_5$	$\bar{x}_3\bar{x}_4\bar{x}_5$
x_1x_2	1	1						
$x_1\bar{x}_2$								
$\bar{x}_1x_2$								
$\bar{x}_1\bar{x}_2$								

b)

	$x_3x_4x_5$	$x_3x_4\bar{x}_5$	$x_3\bar{x}_4x_5$	$x_3\bar{x}_4\bar{x}_5$	$\bar{x}_3x_4x_5$	$\bar{x}_3x_4\bar{x}_5$	$\bar{x}_3\bar{x}_4x_5$	$\bar{x}_3\bar{x}_4\bar{x}_5$
x_1x_2								
$x_1\bar{x}_2$								
$\bar{x}_1x_2$	1			1				
$\bar{x}_1\bar{x}_2$	1			1				

c)

	$x_3x_4x_5$	$x_3x_4\bar{x}_5$	$x_3\bar{x}_4x_5$	$x_3\bar{x}_4\bar{x}_5$	$\bar{x}_3x_4x_5$	$\bar{x}_3x_4\bar{x}_5$	$\bar{x}_3\bar{x}_4x_5$	$\bar{x}_3\bar{x}_4\bar{x}_5$
x_1x_2	1	1					1	1
$x_1\bar{x}_2$								
$\bar{x}_1x_2$								
$\bar{x}_1\bar{x}_2$	1	1					1	1

d)

	$x_3x_4x_5$	$x_3x_4\bar{x}_5$	$x_3\bar{x}_4x_5$	$x_3\bar{x}_4\bar{x}_5$	$\bar{x}_3x_4x_5$	$\bar{x}_3x_4\bar{x}_5$	$\bar{x}_3\bar{x}_4x_5$	$\bar{x}_3\bar{x}_4\bar{x}_5$
x_1x_2					1	1		
$x_1\bar{x}_2$					1	1		
$\bar{x}_1x_2$					1	1		
$\bar{x}_1\bar{x}_2$					1	1		

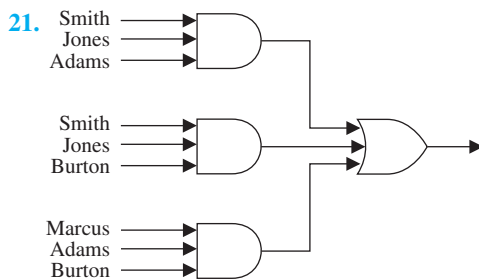
e)

	$x_3x_4x_5$	$x_3x_4\bar{x}_5$	$x_3\bar{x}_4x_5$	$x_3\bar{x}_4\bar{x}_5$	$\bar{x}_3x_4x_5$	$\bar{x}_3x_4\bar{x}_5$	$\bar{x}_3\bar{x}_4x_5$	$\bar{x}_3\bar{x}_4\bar{x}_5$
x_1x_2	1	1	1	1				
$x_1\bar{x}_2$	1	1	1	1				
$\bar{x}_1x_2$	1	1	1	1				
$\bar{x}_1\bar{x}_2$	1	1	1	1				

f)

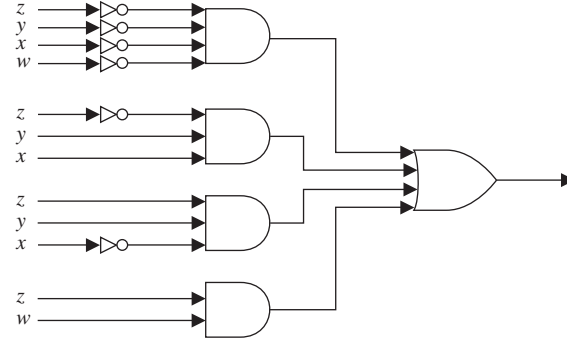
	$x_3x_4x_5$	$x_3x_4\bar{x}_5$	$x_3\bar{x}_4x_5$	$x_3\bar{x}_4\bar{x}_5$	$\bar{x}_3x_4x_5$	$\bar{x}_3x_4\bar{x}_5$	$\bar{x}_3\bar{x}_4x_5$	$\bar{x}_3\bar{x}_4\bar{x}_5$
x_1x_2		1	1			1	1	
$x_1\bar{x}_2$		1	1			1	1	
$\bar{x}_1x_2$		1	1			1	1	
$\bar{x}_1\bar{x}_2$		1	1			1	1	

17. a) 64 b) 6 19. Rows 1 and 4 are considered adjacent. The pairs of columns considered adjacent are: columns 1 and 4, 1 and 12, 1 and 16, 2 and 11, 2 and 15, 3 and 6, 3 and 10, 4 and 9, 5 and 8, 5 and 16, 6 and 15, 7 and 10, 7 and 14, 8 and 13, 9 and 12, 11 and 14, 13 and 16.



23. a) $\bar{x}z$ b) y c) $x\bar{z} + \bar{x}z + \bar{y}z$ d) $xz + \bar{x}y + \bar{y}z$ 25. a) $wxz + w\bar{x}y + w\bar{y}z + w\bar{x}y\bar{z}$ b) $x\bar{y}z + \bar{w}\bar{y}z + w\bar{x}y\bar{z} + w\bar{x}y\bar{z} + \bar{w}\bar{x}y\bar{z}$ c) $\bar{y}z + wxy + w\bar{x}\bar{y} + \bar{w}\bar{x}y\bar{z}$ d) $wy + yz + \bar{x}y + wxz + \bar{w}\bar{x}z$ 27. $x(y + z)$

29.



31. $\bar{x}\bar{z} + xz$ 33. We use induction on n . If $n = 1$, then we are looking at a line segment, labeled 0 at one end and 1 at the other end. The only possible value of k is also 1, and if the literal is x_1 , then the subcube we have is the 0-dimensional subcube consisting of the endpoint labeled 1, and if the literal is $\bar{x}_1$, then the subcube we have is the 0-dimensional subcube consisting of the endpoint labeled 0. Now assume that the statement is true for n ; we must show that it is true for $n + 1$. If the literal x_{n+1} (or its complement) is not part of the product, then by the inductive hypothesis, the product when viewed in the setting of n variables corresponds to an $(n - k)$ -dimensional subcube of the n -dimensional cube, and the Cartesian product of that subcube with the line segment $[0, 1]$ gives us a subcube one dimension higher in our given $(n + 1)$ -dimensional cube, namely having dimension $(n + 1) - k$, as desired. On the other hand, if the literal x_{n+1} (or its complement) is part of the product, then the product of the remaining $k - 1$ literals corresponds to a subcube of dimension $n - (k - 1) = (n + 1) - k$ in the n -dimensional cube, and that slice, at either the 1-end or the 0-end in the last variable, is the desired subcube.

Supplementary Exercises

1. a) $x = 0, y = 0, z = 0; x = 1, y = 1, z = 1$ b) $x = 0, y = 0, z = 0; x = 0, y = 0, z = 1; x = 0, y = 1, z = 0; x = 1, y = 0, z = 1; x = 1, y = 1, z = 0; x = 1, y = 1, z = 1$ c) No values 3. a) Yes b) No c) No d) Yes 5. $2^{2^{n-1}}$ 7. a) If $F(x_1, \dots, x_n) = 1$, then $(F + G)(x_1, \dots, x_n) = F(x_1, \dots, x_n) + G(x_1, \dots, x_n) = 1$ by the dominance law. Hence, $F \leq F + G$. b) If $(FG)(x_1, \dots, x_n) = 1$, then $F(x_1, \dots, x_n) \cdot G(x_1, \dots, x_n) = 1$. Hence, $F(x_1, \dots, x_n) = 1$. It follows that $FG \leq F$. 9. Because $F(x_1, \dots, x_n) = 1$ implies that $F(x_1, \dots, x_n) = 1$, $\leq$ is reflexive. Suppose that $F \leq G$ and $G \leq F$. Then $F(x_1, \dots, x_n) = 1$ if and only if $G(x_1, \dots, x_n) = 1$. This implies that $F = G$. Hence, $\leq$ is antisymmetric. Suppose that $F \leq G \leq H$. Then if $F(x_1, \dots, x_n) = 1$, it follows that $G(x_1, \dots, x_n) = 1$, which implies that $H(x_1, \dots, x_n) = 1$. Hence, $F \leq H$, so $\leq$ is transitive. 11. a) $x = 1, y = 0, z = 0$ b) $x = 1, y = 0, z = 0$ c) $x = 1, y = 0, z = 0$